1. Identity

Document	STAR-PKG-STAR-SLERP-EN · v1.3 · 2026-10-16
Package	star-slerp
Version	0.1.0
Capability	star.attitude.slerp
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_slerp

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_slerp-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_slerp-0.1.0-py3-none-any.whl	d5ba548c3f09edbd4a7ab6c70777b86e17620ab55744744f1308a07412203cd3
star_slerp.py	74c8063585c8506e6ebd51060392d9af2b18fc5f83ef5be6d28a689efaabb6be
star_slerp-0.1.0-source.zip	6d14996805b2ed0470b5fde1eea7b3e5a694a40f5dee1748c8a577796e0fb20b

5. What it does, and its limits

Spherical linear interpolation between unit quaternions along the shorter rotation, and interpolation in a table of attitudes at given times; on 600 pairs of attitudes the largest component difference is 4.4e-16 from SciPy Slerp and 3.3e-16 from a route through NAIF SPICE rotation matrices, axis and angle.

Interpolation at constant rate between two attitudes, nothing else: no angular velocity, no smooth (C1) interpolation across table rows (the rate jumps at each row), no extrapolation, no model of the true motion between samples. For a rotation of exactly 180 degrees both arcs are equally short and the result follows the sign of the q1 supplied; other libraries may choose the other arc. The quaternion convention is scalar-first Hamilton: data in another convention must be converted before use.

- `slerp(q0, q1, t)` returns the attitude reached after the fraction t of the rotation from q0 to q1, about the same axis and at constant rate; quaternions are (w, x, y, z), scalar first, of unit length.
- The path is always the shorter rotation: when q0 and q1 are in opposite hemispheres q1 is negated first. The result is of unit length and in the hemisphere of q0. `interpolate(times, quaternions, t)` applies the same rule between the two rows of the table that bracket t and refuses a t outside the table.
- Measured on 600 pairs of attitudes (rotations from 1e-9 rad to within 0.1 degree of 180 degrees, either hemisphere): the largest component difference is 4.4e-16 from SciPy `Slerp`, 3.3e-16 from a route through NAIF SPICE rotation matrices, axis and angle, and 2.2e-16 from ahrs above its linear-interpolation threshold.
- Every function returns a tuple of four floats or raises `ValueError`: a quaternion that is not four finite numbers of unit length within 1e-9 (never normalised silently), t out of range, a malformed or non-increasing table.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_slerp_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-slerp}: Interpolation of attitudes along the shortest arc},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_slerp/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts